

Solution Requirements Specifications

Functional and Technical Matrix

1. Functional Requirements

- **User Management:** Patients can sign up, update profiles, and manage history. Doctors can detail medical criteria. Admins check system metrics.
- **Specialist Directory:** Sort and locate active professionals. Dynamic pagination for fast rendering.
- **Appointment Life Cycle:** Pending reservations -> Admin check/Doctor approval -> Digital completion and prescription write-up.
- **Notifications Handler:** System broadcasts status changes in real-time to the target user's active screen.

2. Non-Functional and Technical Requirements

- **Performance:** Pages must fetch API content in <1.5 seconds under standard Wi-Fi conditions. Use page pagination and selective indexing.
- **Security and Privacy:** All user passwords must be hashed before storage. Session routes secured via JWT validation. Exclude cross-origin cookie sharing.
- **Network Compatibility:** Allow local network sharing using binding techniques so users on the same subnet can use the system on their laptops.